

100 Days of Linux - Day 051

Title: Testing Network Connectivity with ping

Today's Goal

Learn how to test network connectivity and measure response time using the **ping** command.

Why This Matters

Before accessing servers, APIs, databases, or cloud resources, engineers often verify that a machine is reachable. Ping is one of the first troubleshooting commands used on Linux systems.

Command of the Day: ping

Syntax:

```
ping hostname  
ping IP_ADDRESS  
ping -c 4 google.com (send only 4 packets)
```

Common Options:

- c = Number of packets to send
- i = Interval between packets
- W = Timeout for each reply

Important Output:

icmp_seq, ttl, time, packet loss, min/avg/max response time

Beginner Lesson

The ping command sends small network packets to another computer and waits for replies. If replies are received, the destination is reachable. High response times or packet loss may indicate network problems.

Practice Questions

1. Ping localhost four times using the appropriate option.
2. Ping google.com four times and observe the response time.
3. Ping 8.8.8.8 four times and compare the results with google.com.
4. Run ping to a hostname that does not exist and observe the error message.
5. Display your current working directory after completing today's tasks.

Hints

Use the -c option so ping stops automatically after sending a fixed number of packets.

Mini Challenge

Ping localhost, 8.8.8.8, and google.com. Record the average response time for each and identify which responds the fastest.

Common Mistakes

Forgetting that ping runs continuously by default on Linux. Press Ctrl+C to stop it if you don't use -c.

Did You Know?

Some servers intentionally block ICMP echo requests, so a failed ping does not always mean the server is offline.

Pro Tip

When troubleshooting connectivity, test localhost first, then your gateway, then a public IP, and finally a hostname to narrow down the problem.

Answer Key (Instructor Only)

Q1: ping -c 4 localhost

Q2: ping -c 4 google.com

Q3: ping -c 4 8.8.8.8

Q4: ping -c 4 invalid-hostname

Q5: pwd

Homework

Practice pinging different websites and IP addresses. Compare response times and note any packet loss.

Next Day Preview

Tomorrow you'll learn the curl command to retrieve web pages, test REST APIs, and download data from the command line.